Electrical and Servicing Systems

Eduardo Manuel Godinho Rodrigues

上海大学里斯本学院
ULISBOA SCHOOL, SHANGHAI UNIVERSITY
ULisboa School, Shanghai University

Electrical and Servicing Systems

Protection Devices

Electrical installation protection devices safeguard against overloads, short circuits, and surges, including fuses, circuit breakers (MCBs, MCCBs, and RCBOs), residual current devices (RCDs/RCCBs), and surge protective devices (SPDs)

Circuit breakers automatically cut off power to prevent damage from overloads or short circuits by interrupting the current flow when a fault is detected, acting as a safety device.

3P+N
(high rated current Ia)

3P+N
(low rated current Ia)

P+N

P

Electrical and Servicing Systems

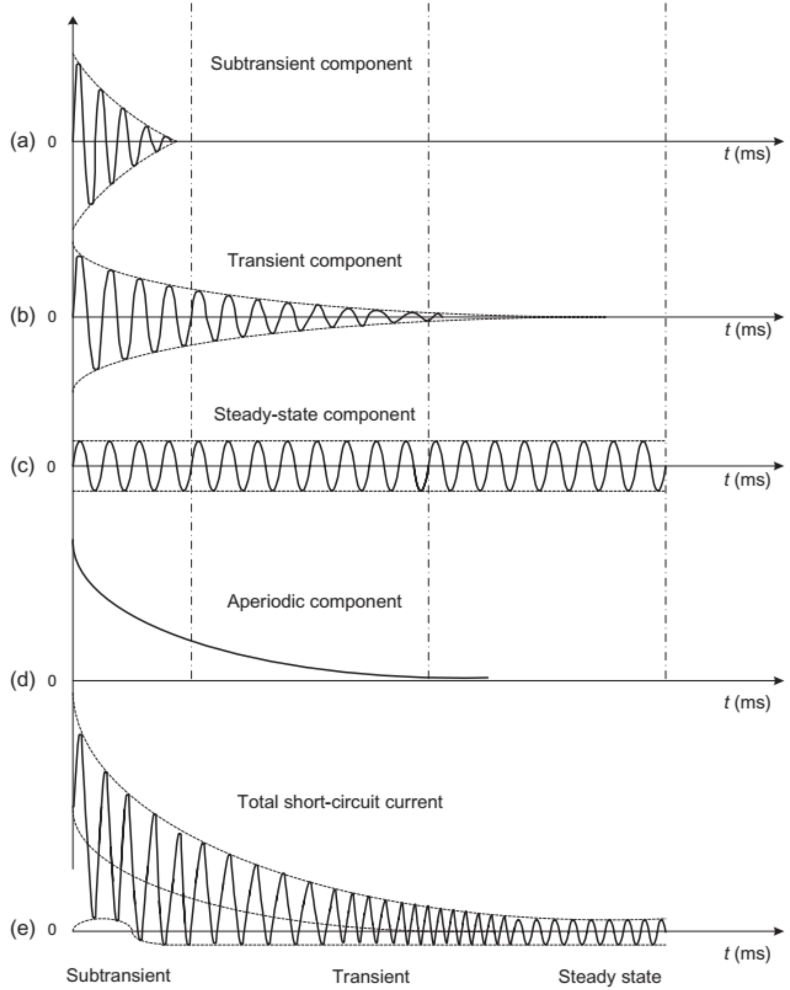
Protection Devices

Residual Current Circuit Breakers (RCCB).

Electrical and Servicing Systems

Protection Devices

Short-circuit current waveform



Components of generator short-circuit current.

Electrical and Servicing Systems

Protection Devices

Ics

It is the rated service short-circuit breaking capacity

Icu

The ultimate **rated breaking capacity** (Icu) or (Icn) is the maximum fault-current a circuit-breaker can successfully interrupt without being damaged. These are very high currents of extremely low probability. In normal circumstances, the fault-currents are considerably less than the (Icu) of the circuit breaker.

In a correctly designed installation, a circuit-breaker is never required to operate at its maximum breaking current (Icu). For this reason, a new characteristic has been introduced which is the rated service short-circuit breaking capacity (Ics). IEC 60947-2 expresses (Ics) as a percentage (25, 50, 75, 100%) of (Icu)

Electrical and Servicing Systems

Protection Devices

This Ics is the value of the short-circuit threshold current. The device trips and after resetting, guarantees the installation's continuity of service.

If the short-circuit current reaches the value Icu, the device will trip safely and after resetting, there is no guarantee that the installation will continue to operate.

EN 60-898

IEC 947-2

IEC 60947-2

Electrical and Servicing Systems

Protection Devices

EN 60-898 IEC 947-2

Defines the rated breaking capacity Ics for use by unauthorized persons 5 consecutive resets.

Ics → EN60898 , 4.5 kA

Icu → EN 60947-2 , 6 kA

EN 60-898 IEC 947-2

4,5 / 6kA

Electrical and Servicing Systems

Protection Devices

EN 60-898 IEC 947-2

Defines the rated breaking capacity Ics for use by unauthorized persons 5 consecutive resets.

EN60898 , 50% Pdc ultimo

Ics → EN 60947-2 , 50 kA

Icu → EN 60-898 IEC 60472-2

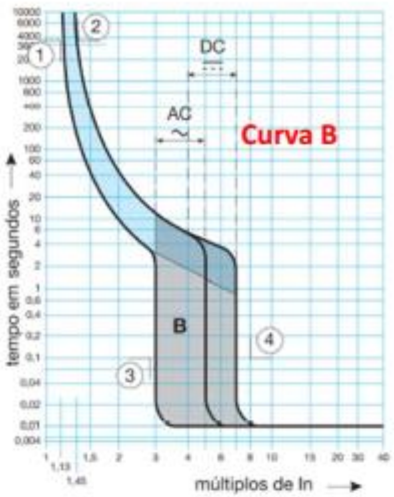
25 / 50kA

Electrical and Servicing Systems

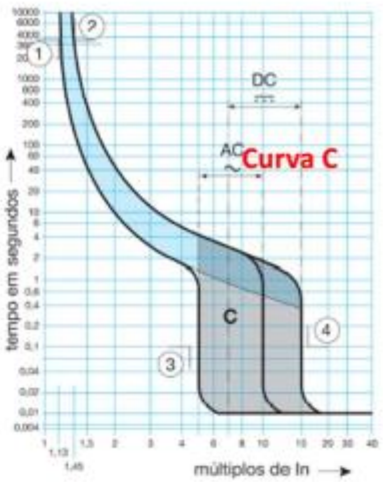
Protection Devices

Circuit-breaker magnetic tripping curves

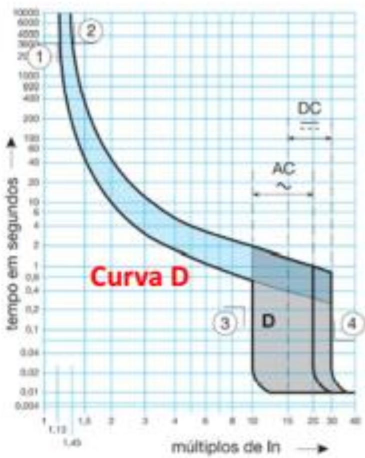
Curve B



Curve C



Curve D

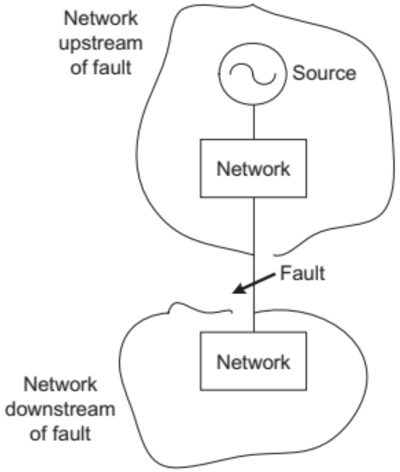


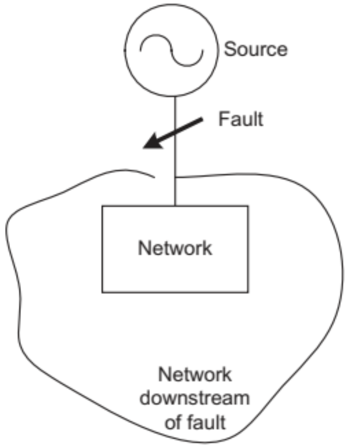
Quadro 2 disparo		curva B AC ~ 50 Hz	curva B DC ⎓	curva C AC ~ 50 Hz	curva C DC ⎓	curva D AC ~ 50 Hz	curva D DC ⎓
①	I_{t1}	1,13 In	1,13 In	1,13 In	1,13 In	1,13 In	1,13 In
②	I_{t2}	1,45 In	1,45 In	1,45 In	1,45 In	1,45 In	1,45 In
③	I_{rm1}	3 In	4 In	5 In	7 In	10 In	15 In
④	I_{rm2}	5 In	7 In	10 In	15 In	20 In	30 In

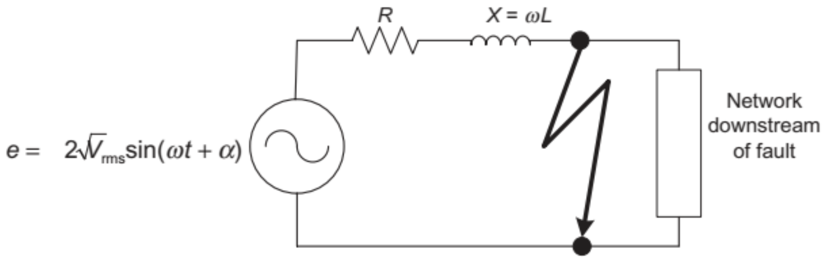
Electrical and Servicing Systems

Short-circuit calculation

The short - circuit current is highly dependent on the equivalent impedance of the network seen at the fault point. The value of this impedance depends on the network configuration, type of system earthing, elements included in the network, type of fault, and fault location







Fault location far from the source

Fault location at generator terminals

Electrical and Servicing Systems

Short-circuit calculation

The method for calculating short-circuit currents should be chosen according to:

- Currents to be calculated (maximum short-circuit, minimum short-circuit);
- Degree of accuracy in the approximate calculation of short-circuit currents;
- Known characteristics of the supply and the various parameters of the electrical installation;
- Importance of the installation;

The most used calculation methods are:

a) Composition method;
b) Conventional method;
c) Impedance method.

Electrical and Servicing Systems

Short-circuit calculation

Composition method

This method is a simplified approach, used when the characteristics of the power supply are unknown.

The upstream short-circuit impedance is calculated based on an estimated value of the short-circuit current at the source.

By knowing the symmetrical short-circuit current at the origin of the installation, it is possible to estimate the presumed maximum short-circuit current at the end of the circuit with a certain known length and section.

The maximum short-circuit current at any point in the installation is determined using tables, based on knowledge of the value of the assumed short-circuit current at the origin of the installation, the length of the line, and the nature and cross-section of the conductors, This method applies to installations whose power does not exceed 800 kVA.

Electrical and Servicing Systems

Short-circuit calculation

Conventional method

This method makes it possible to determine the minimum short-circuit currents and the fault current at the end of the circuit without needing to know the characteristics of the installation upstream of the circuit in question.

The method is specified in the French standard UTE C 15 - 105:2003 and is based on the assumption that the voltage at the origin of the circuit is equal to 80 per cent of the nominal voltage of the installation during the short-circuit or fault under consideration.

The minimum short-circuit currents are obtained from the expression:

$$I_{K1Min} = \frac{0.8 * 1.05 * U_o}{Z_F + Z_N}$$

I_{K1Min} minimum short-circuit current (A)

U_o nominal voltage between phase and neutral (V)

Z_F phase conductor resistance (Ω)

Z_N neutral conductor resistance (Ω)

Electrical and Servicing Systems

Short-circuit calculation

Conventional method

The fault current at the installation end is obtained from the expression:

$$I_d = \frac{0.8 * 1.05 * U_o}{Z_F + Z_{PE}}$$

I_d fault current (A)

U_o nominal voltage between phase and neutral (V)

Z_F phase conductor resistance (Ω)

Z_{PE} protective conductor (Ω)

Electrical and Servicing Systems

Short-circuit calculation

Impedance method

The impedance method is based on determining the short-circuit currents from the fault impedance that represents the loop through which the fault current is travelling.

It consists of separately adding up the various resistances and reactances of the fault loop from the power source to the point where the fault is considered and calculating the corresponding impedance.

It makes it possible to calculate all short-circuit currents (maximum, minimum, three-phase, two-phase, single-phase) and fault currents at all points in an installation with good accuracy, provided that all the characteristics of the fault loop are known.

Electrical and Servicing Systems

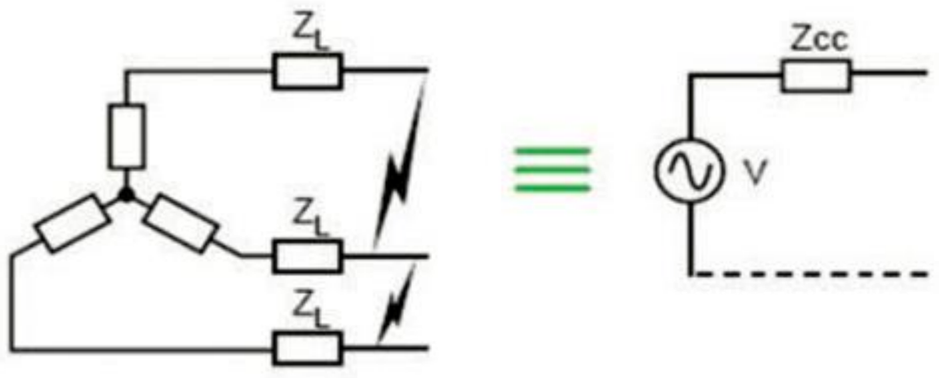
Short-circuit calculation

Maximum currents

The maximum short-circuit current will be calculated by simulating the occurrence of such a fault at the point where the protection device is installed.

Symmetrical three-phase short-circuit current (lk3 Max)

This type of fault is the most severe fault as it produces the highest short - circuit current with equal magnitudes in each of the three phases.



Electrical and Servicing Systems

Short-circuit calculation

Maximum currents

Symmetrical three-phase short-circuit current (lk3 Max)

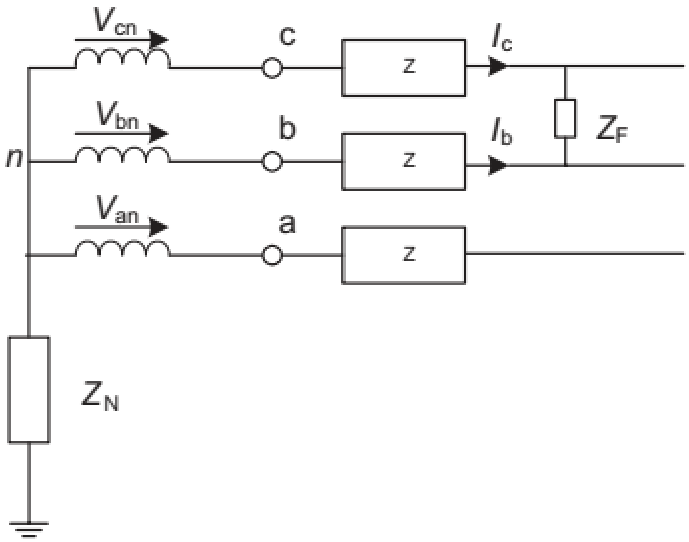
R_Q, X_Q	Resistance and reactance upstream of the transformer (Ω)
R_T, X_T	Resistance and reactance of the transformer (Ω)
R_{Uph}, X_{Uph}	Resistance and reactance of the phase conductor from the transformer to the origin of the circuit in analysis (Ω)
L	Length of the pipe (m)
S	Cross-section of the conductors in the circuit in analysis(mm2)
n_{ph}	Number of conductors in parallel per phase
ρ_0	Resistivity of the conductors at 20 °C (Qxmm2/m)
λ	Reactance per unit length of the conductors
U_o	Nominal voltage between phase and neutral (V)

$$I_{k3Máx.} = \frac{C_{máx.} \times m \times U_o}{\sqrt{\left[R_Q + R_T + R_{Uph} + \rho_0 \frac{L}{S_{n\,ph}}\right]^2 + \left[X_Q + X_T + X_{Uph} + \lambda \times \frac{L}{n_{ph}}\right]^2}}$$

Electrical and Servicing Systems

Short-circuit calculation

Two-phase short-circuit current



$$I_{k2Máx.} = \frac{\sqrt{3}}{2} \times I_{k3Máx.} = 0,86 \times I_{k3Máx.}$$

I_K short-circuit current (A)

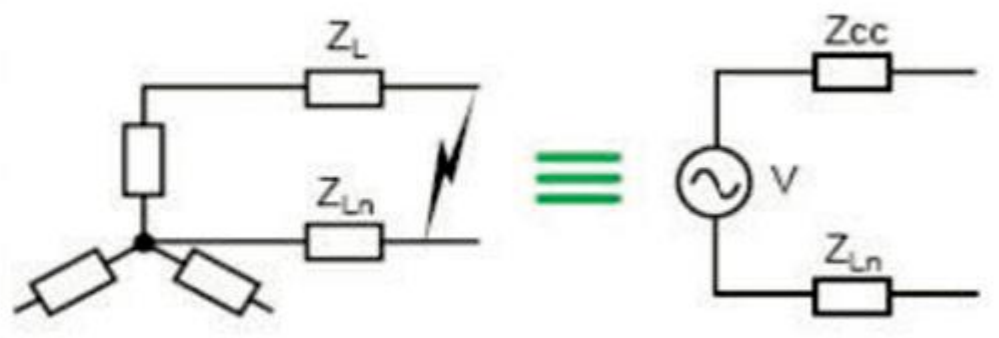
I_{k2Max} two phase short-circuit current (A)

I_{k3Max} symmetric short-circuit current (A)

Electrical and Servicing Systems

Short-circuit calculation

Phase to neutral short-circuit current



$$I_{k1Máx.}=\frac{C_{máx.}\times m\times U_o}{\sqrt{\left[R_Q+R_T+R_{Uph}+R_{UN}+\rho_0\times L\times\left(\frac{1}{S_{n\,ph}}+\frac{1}{S_N\times S_N}\right)\right]^2+\left[X_Q+X_T+X_{Uph}+X_{UN}+\lambda\times L\times\left(\frac{1}{n_{ph}}+\frac{1}{n_N}\right)\right]^2}}$$

Electrical and Servicing Systems

Short-circuit calculation

Phase to neutral short-circuit current

$I_{k1\,Máx}$	Short-circuit current circuit (A)
R_Q, X_Q	Resistance and reactance upstream of the transformer (Ω)
R_T, X_T	Resistance and reactance of the transformer (Ω)
R_{Uph}, X_{Uph}	Resistance and reactance of the phase conductor from the transformer to the origin of the circuit under analysis (Ω)
R_{UN}, X_{UN}	Resistance and reactance of the neutral conductor from the transformer to the origin of the circuit under analysis ()
L	Length of the conductors (m)
S	Cross-section of line conductors of the circuit under analysis (mm2)
S_N	Cross-section of the neutral conductor of the circuit under analysis (mm2)
n_{ph}	Number of conductors in parallel per phase
n_N	Number of conductors in parallel in the neutral conductor
ρ_0	Resistivity of the conductors at 20 °C (xmm2/m)
λ	Reactance per unit length of the conductors
U_0	Nominal voltage between phase and neutral (V)

Electrical and Servicing Systems

Short-circuit calculation

Minimum currents

The minimum short-circuit current will be calculated by simulating the occurrence of a short-circuit at the furthest point of the circuit that the protection device is protecting.

a) Three-phase circuit without neutral

Calculated using the same formula as the maximum two-phase short-circuit current (I_{k2max}), but in which the resistivity of the conductors at 20 °C (ρ_0) is replaced by the resistivity $\rho_1(\rho_1 = 1.25(\rho_0)$, if circuit breakers are used, and ρ_2 ($\rho_2 = 1.5\rho_0$), if fuses are used, and Cmax is replaced by Cmin.

b) Three-phase circuit with neutral or single-phase phase/neutral

Calculated using the same formula as the maximum two-phase short-circuit current (I_{k1max}), but in which the resistivity of the conductors at 20 °C (ρ_0) is replaced by the resistivity $\rho_1(\rho_1 = 1.25(\rho_0)$ if circuit breakers are used, and ρ_2 ($\rho_2 = 1.5\rho_0$), if fuses are used, and Cmax is replaced by Cmin.

Electrical and Servicing Systems

Short-circuit impedance calculation

Impedance of the upstream network

With knowledge of the upstream short-circuit power, it is possible to determine the equivalent impedance of this network:

$$Z_Q = \frac{(m \times U_n)^2}{S_{kQ}}$$

Z_q network upstream impedance (Ω)

S_{kQ} short-circuit power of MV side (MVA)

m load factor (equal to 1.05)

V_n phase to phase voltage (V)

Transformer substation location	S_{kQ} (MVA)
Rural area	150
Semi-urban area	250
Urban area	350-500

The average S_{kQ} values on the MV side at transformer substation should be obtained from the power utility.

Electrical and Servicing Systems

Short-circuit impedance calculation

Transformer impedance (Zt)

Calculated from the short-circuit voltage (U), expressed in %.

$$Z_T = \frac{U_{kr}}{100} \times \frac{(m \times U_n)^2}{S_{rT}}$$

m load factor (equal to 1.05)

V_n phase to phase voltage (V)

U_{kt} short-circuit voltage (%)

Z_t transformer impedance (Ω)

S_{rt} transformer rated power (kW)

Short-circuit voltage as function of MV transformer apparent power

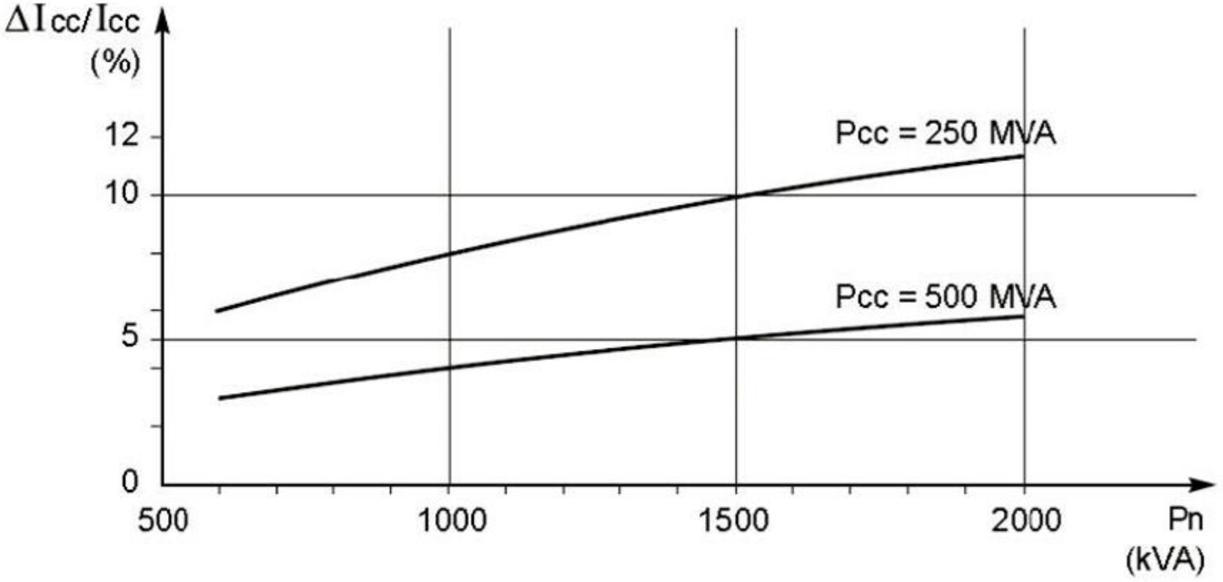
Aparente Power (kVA)	≤630	800	1000	1250	1600	2000
Short-circuit voltage (%)	4	5	5	5	6	6

The EN 60076-5140 standard defines the short-circuit voltages for power transformers.

Electrical and Servicing Systems

Short-circuit impedance calculation

Transformer impedance (Zt)



It is clear that this impedance can be disregarded for networks where the short-circuit power (Scc) is much higher than the rated power (Sn) of the transformer.

Electrical and Servicing Systems

Short-circuit impedance calculation

Impedance of the alternator or synchronous machine

Due to their high internal impedance, synchronous machines generate much lower short-circuit currents than transformers of equivalent power.

The development of a short-circuit current at the terminals of an alternator can be divided into three periods:

- sub-transient period: from 10 to 20 ms, during which the level of the short-circuit current is the highest (> 5 In);
- transient period: up to 200 to 300 ms, during which the level of the short-circuit current is in the order of 3 to 5 In;
- the short-circuit current stabilises at a level ranging from 0.3 to 5 In, depending on the type of alternator excitation.

Electrical and Servicing Systems

Short-circuit impedance calculation

Impedance of the alternator or synchronous machine

The impedance of the alternator or synchronous machine is calculated from knowledge of the alternator's reactance, expressed in %..

$$X'_d = \frac{X'_d}{100} \times \frac{(U_n)^2}{S_{rG}}$$

$$X_0 = \frac{X_0}{100} \times \frac{(U_n)^2}{S_{rG}}$$

Z_t transformer impedance (Ω)

X'_d homopolar reactance (%)

X_o homopolar reactance (%)

V_o phase to neutral (V)

S_{rg} alternator rated power (kVA9

Typical alternator reactance values

	Sub-transient reactance	Transient reactance	Permanent reactance
Turbo alternators	10 a 20	15 a 25	150 a 230
Pole-surface alternators	15 a 25	25 a 35	70 a 120

Electrical and Servicing Systems

Short-circuit impedance calculation

Impedance of conductors and cables

The impedance of conductors and cables depends on their resistance and reactance.

$$R = \rho \times \cancel{\theta} \frac{L}{n_c \times S_c}$$

$$X = \lambda \frac{L}{n_c}$$

R	resistance (mΩ)
ρ	conductor resistivity (Ωmm2/m)
n_c	number of conductors in parallel
S_c	cross-section (mm2)
L	length (m)
X	reactance (mΩ)
λ	conductor linear reactance (mΩ/m)
n_c	number of conductors in parallel
L	length (m)

In practice, at low voltage and for conductors with a cross-section of less than 150 mm2, only the resistance value is taken into account (RL < 0.15 mN/m, where S > 150 mm2).

Electrical and Servicing Systems

Short-circuit impedance calculation

Impedance of conductors and cables

Resistivity values of the conductors used to consider the maximum and minimum short-circuit current calculated.

Rule	Resistivity	Copper resistivity	Aluminium resistivity	Electrical conductors involved
Maximum short-circuit current	$\rho_1 = 1.25\rho_{20}$	0.0225	0.036	Phase/Neutral
Minimum short-circuit current	$\rho_1 = 1.5\rho_{20}$	0.027	0.043	Phase/Neutral
TN and IT fault current	$\rho_1 = 1.25\rho_{20}$	0.0225	0.036	Phase/Neutral PE/PEN

ρ_{20} conductor resistivity at 20 °C:

0.018 Ωmm2/m for copper and 0.029 Ωmm2/m for aluminium.